FACULTY OF COMPUTING

SEMESTER 3 2024/2025

SECP3723 SYSTEM DEVELOPMENT TECHNOLGY

Individual Project

Course Registration System

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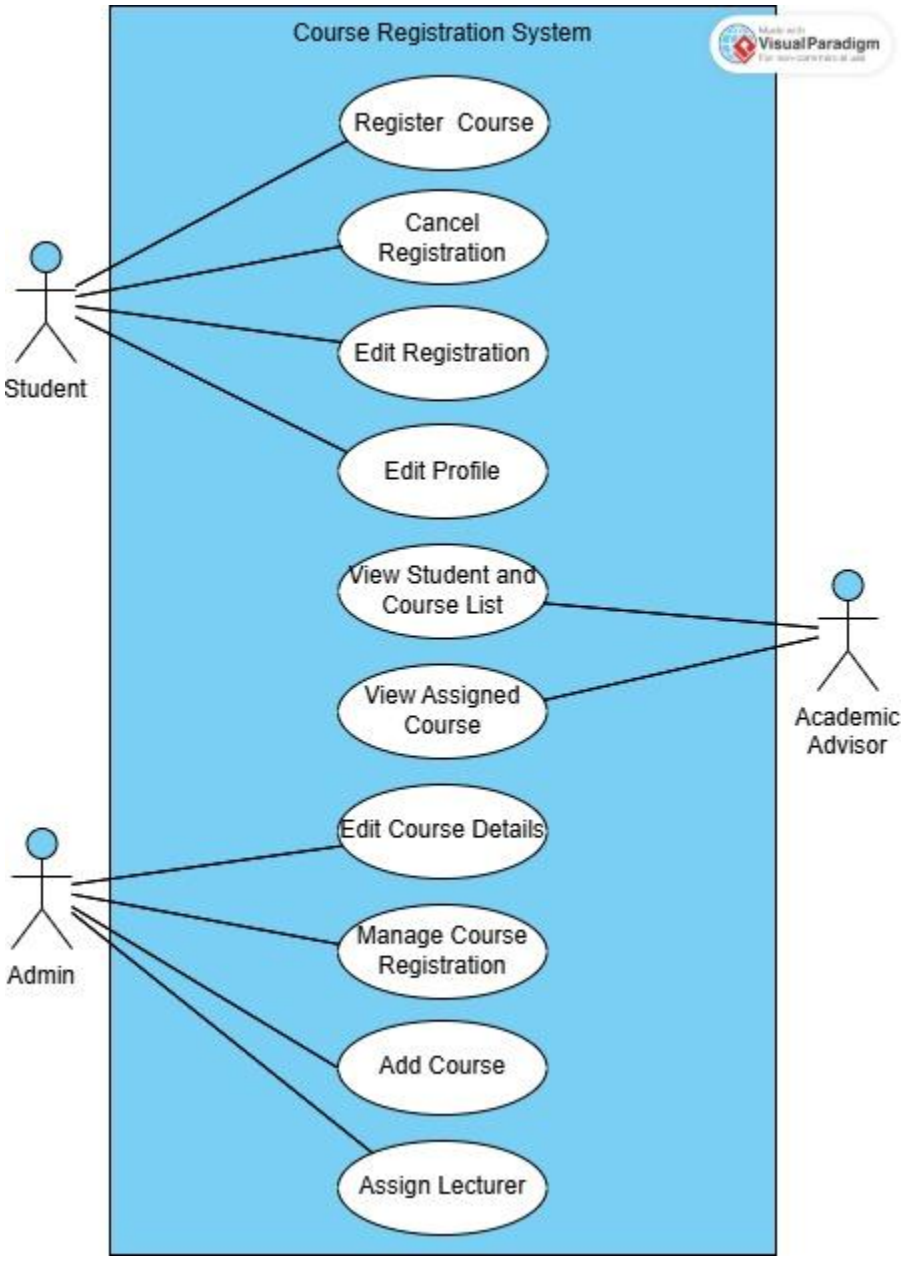
1.0 System Introduction

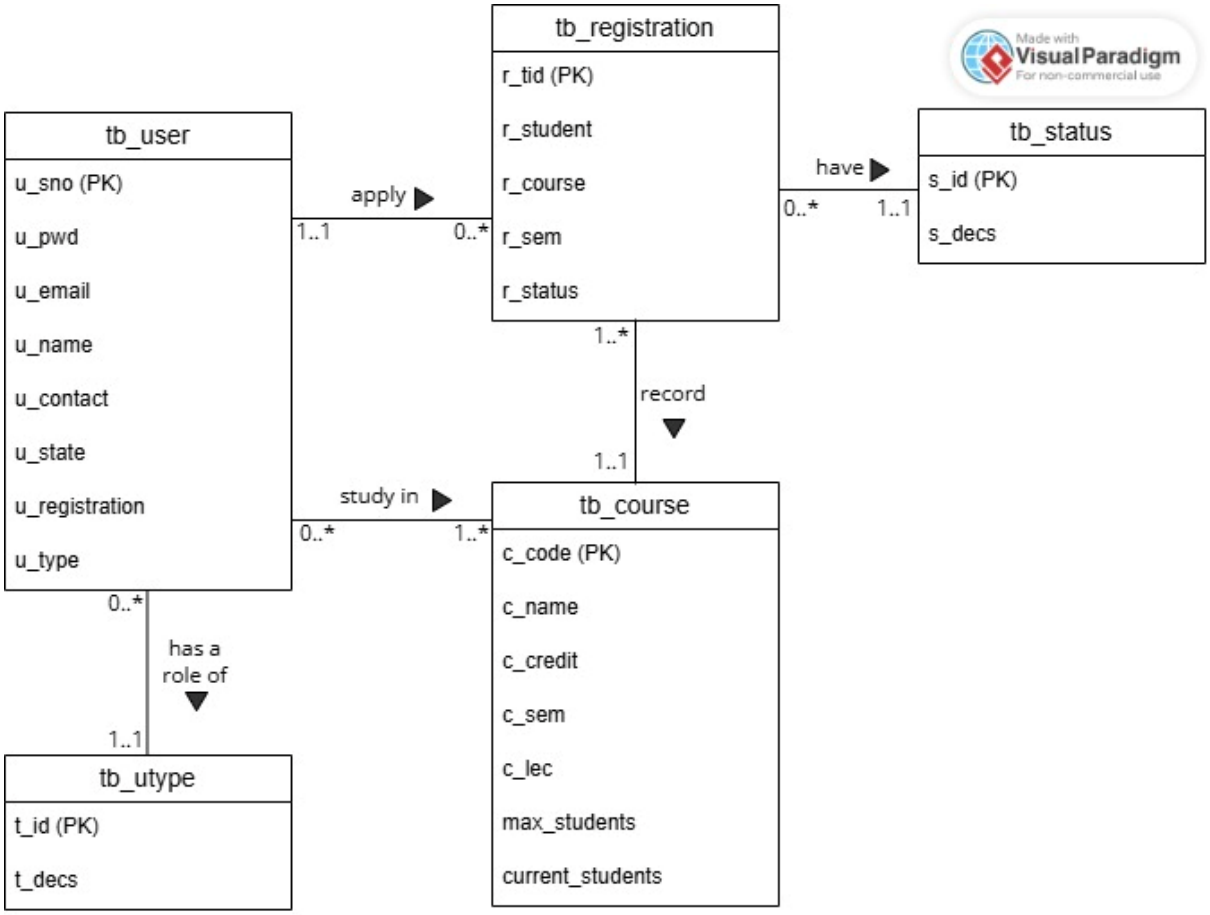
A web-based platform known as the Course Registration System was developed to improve student enrollment, lecturer assignments, and course management easier. Administrators can employ it to assign lecturers, manage courses, registration and view all users in the system. While students can register for available courses with automatic approval if class capacity is enough, lecturers can check their allocated courses and student lists. The system, which was developed using PHP, MySQL, and JavaScript and features bootstrap 5, ensures role-based access control, data validation, and an efficient registration procedure, improving user experience when applying for course registration each semester.

2.0 System Design

2.1 Use Case



2.2 ERD


tb_registration
r_tid (PK)
r_student
r_course
r_sem
r_status
tb_user
u_sno (PK)
u_pwd
u_email
u_name
u_contact
u_state
u_registration
u_type
tb_status
s_id (PK)
s_decs
apply
1..1
0..*
have
0..*
1..1
1..*
record
1..1
study in
0..*
1..*
tb_course
c_code (PK)
c_name
c_credit
c_sem
c_lec
max_students
current_students
0..*
has a
role of
1..1
tb_utype
t_id (PK)
t_decs
Made with
Visual Paradigm
For non-commercial use

3.0 Development Software, Language, Technology, and Tools

3.1 Development Software

- Visual Studio Code – As code editor
- XAMPP – Local server environment for PHP development

3.2 Programming Languages

- PHP – Server-side scripting language for web development. Official Site
- JavaScript – Used for client-side scripting. Learn More
- HTML & CSS – For structuring and styling web pages. HTML | CSS

3.3 Database Technology

- MySQL – Relational database management system. Official Site

3.4 Frameworks & Libraries:

- Bootstrap 5 – Use bootstrap theme as website design
- W3school – As libraries for references

3.5 Other Tools

- phpMyAdmin – Database management tool

4.0 Database Design

tb_course

c_code	c_name	c_credit	c_sem	c_lec	max_students	current_students
SECJ1033	Programming Technique 1	3	2024/2025-1	L001	3	2
SECJ1033	Programming Technique 1	3	2024/2025-2	L002	2	1
SECP2523	Database (WBL)	3	2024/2025-1	L002	3	1
SECP3032	Data Engineering	3	2024/2025-2	L001	3	1
SECP3032	Data Engineering	3	2024/2025-3	L002	3	1
SECP3204	Software Engineering (WBL)	3	2024/2025-1	L001	2	1

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**c_code**	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
2	**c_name**	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
3	**c_credit**	int(11)			No	*None*			Change Drop More
4	**c_sem**	varchar(11)	utf8mb4_general_ci		No	*None*			Change Drop More
5	**c_lec**	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
6	**max_students**	int(11)			No	*None*			Change Drop More
7	**current_students**	int(11)			Yes	0			Change Drop More

tb_registration

r_tid This is the transcation id	r_student	r_course	r_sem	r_status
1	S001	SECJ1033	2024/2025-1	2
2	S003	SECJ1033	2024/2025-1	2
3	S003	SECP3032	2024/2025-3	2
4	S002	SECP3204	2024/2025-1	2
5	S002	SECP3032	2024/2025-2	2
6	S004	SECJ1033	2024/2025-2	2
7	S004	SECP2523	2024/2025-1	2

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**r_tid**	int(11)			No	*None*	This is the transcation id	AUTO_INCREMENT	Change Drop More
2	**r_student**	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
3	**r_course**	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
4	**r_sem**	varchar(11)	utf8mb4_general_ci		No	*None*			Change Drop More
5	**r_status**	int(11)			No	*None*			Change Drop More

tb_user

u_sno	u_pwd	u_email	u_name	u_contact	u_state	u_registration	u_type
I001	$2y$10$Xd4OvNi/IsmaPPT8YImdVO/mL7dezbaekDkExSV4vZI...	itstaff@gmail.com	Ahmad bin Abu	123456789	Johor	2025-01-26	3
I002	$2y$10$RadiVBmQZticGFZaFa3soORSQ50ueu0wRspVjzAQZCu...	adam@gmail.com	Adam Abdul	123456789	Johor	2025-01-01	3
L001	$2y$10$hHt4QzMRLnQJ.sBv4p8/B.IzFgH7jM.nlXFSxCW.m3a...	aina@gmail.com	Aina Abdul	191111111	Johor	2023-06-22	1
L002	$2y$10$3eyz26Pp/axzNTCRFc7QYu8cPi5Nb0xDFISJt/gjH3B...	faz@gmail.com	Fazura Abdul	172222222	Kelantan	2024-01-22	1
S001	$2y$10$/sffeOe262m8ILS8hQ.uiuMuDAJ/TT3ARPS7UCq9kEq...	fat@gmail.com	Fatah Abdul	133333333	Melaka	2023-11-08	2
S002	$2y$10$AUkMSGd33cSV.vJpIBkgPOJZ0YWJiGGXmW7DBNzsNyf...	kam@gmail.com	Kamarul Abdul	124444444	Selangor	2024-10-30	2
S003	$2y$10$XOAvTgRkrBO22JSOmFzl.uob8nRGry2Fro9VrcJUuj1...	fatimah@gmail.com	Fatimah Abdul	123456789	Pulau Pinang	2025-01-27	2
S004	$2y$10$PNK6YWTPAXb61GLg7hMP8.wDVPfe9xKaYncCsTK85Qo...	filza@gmail.com	Filza Abdul	167846693	Sabah	2025-01-27	2

Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
u_sno	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
u_pwd	varchar(255)	utf8mb4_general_ci		No	*None*			Change Drop More
u_email	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
u_name	varchar(50)	utf8mb4_general_ci		No	*None*			Change Drop More
u_contact	int(11)			No	*None*			Change Drop More
u_state	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
u_registration	date			Yes	*NULL*			Change Drop More
u_type	int(11)			No	*None*			Change Drop More

tb_status

←T→			s_id	s_decs
Edit	Copy	Delete	1	Received
Edit	Copy	Delete	2	Approved
Edit	Copy	Delete	3	Rejected
Edit	Copy	Delete	4	Cancelled

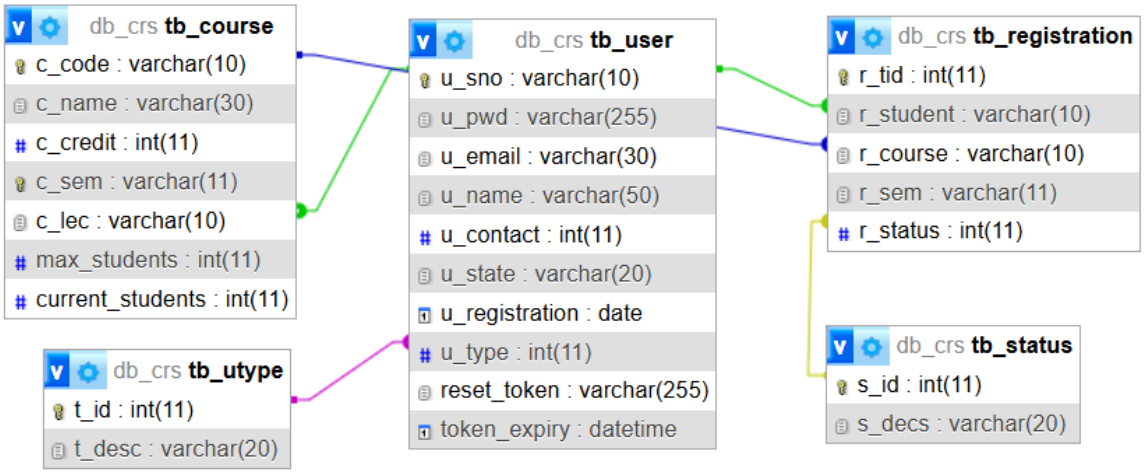
Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
s_id	int(11)			No	*None*			Change Drop More
s_decs	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More

tb_utype

t_id	t_desc
1	Lecturer
2	Student
3	IT Staff

Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
t_id	int(11)			No	*None*			Change Drop More
t_desc	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More

Designer



5.0 Development Step

5.1 Requirement Analysis

- Gather system requirement

5.2 System Design

- Create database scheme and wireframe
- Choose languages and tools to develop, such as PHP, MySQL, Bootstrap, JavaScript

5.3 Implementation (Coding & Development)

- Develop individual modules, with the order of log in, students' module, admins' module and academic advisors' module

5.4 Testing

- Test the functionality of the system
- Identify and fix bugs before deployment

5.5 Deployment

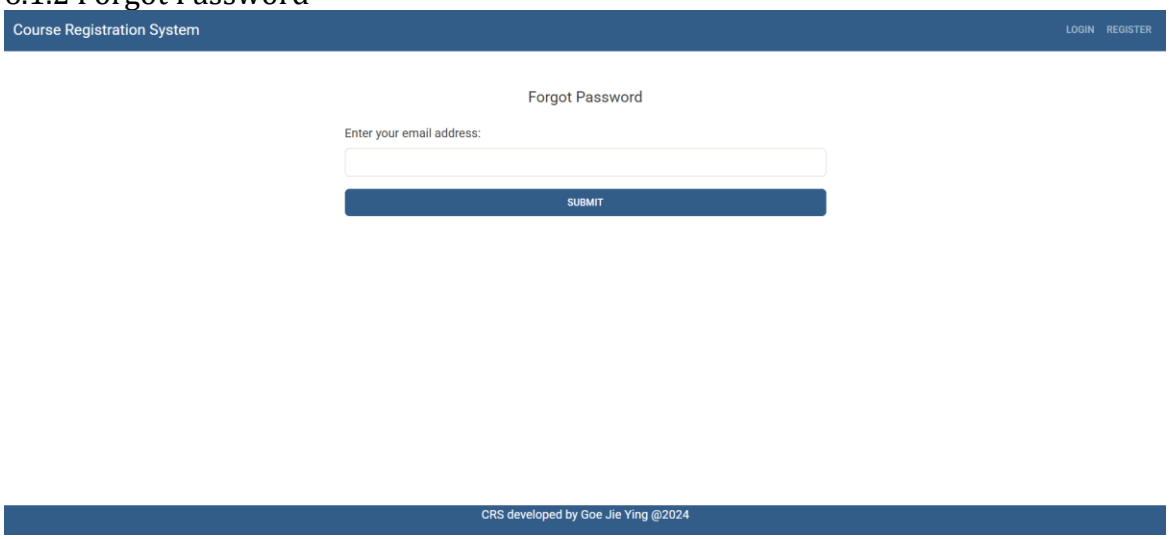
- Deploy the system using InfinityFree Hosting website
- Ensure proper configuration and server setup

6.0 System Interface

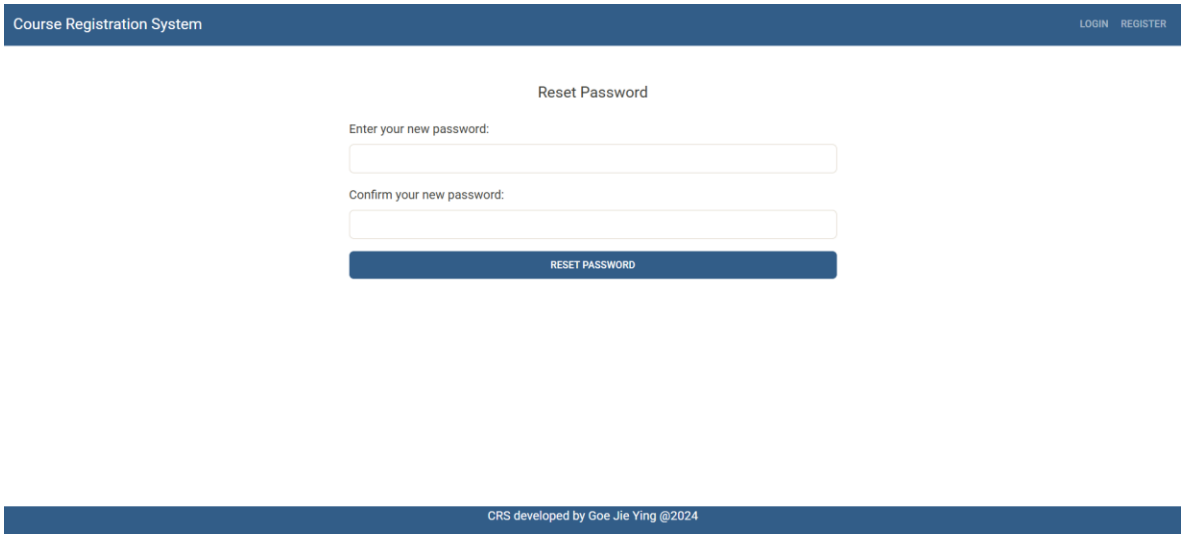
6.1 Log In

6.1.1 Log in Interface

6.1.2 Forgot Password



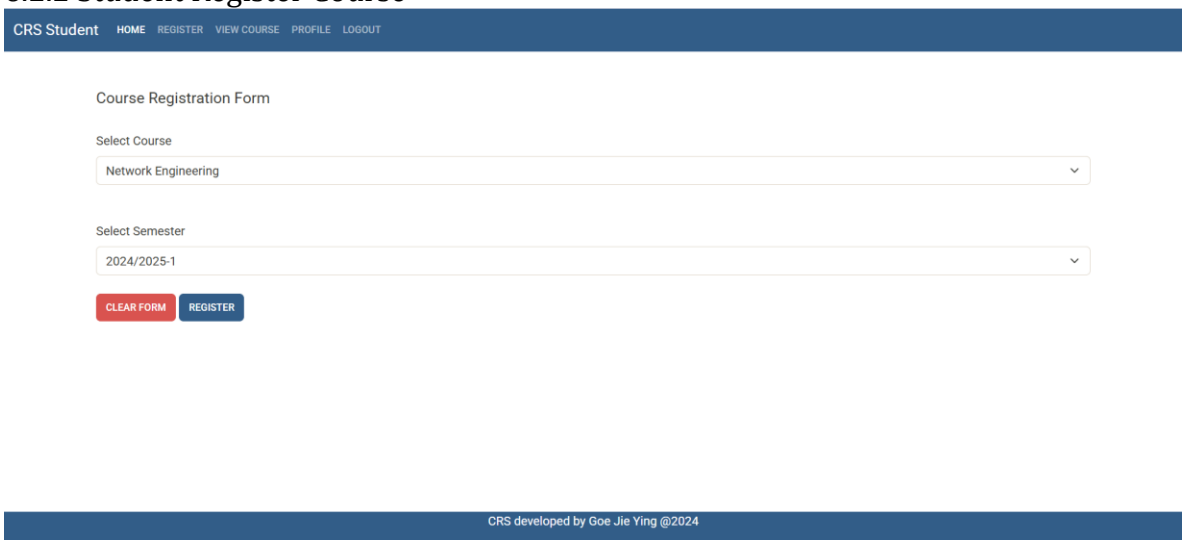
6.1.3 Reset Password



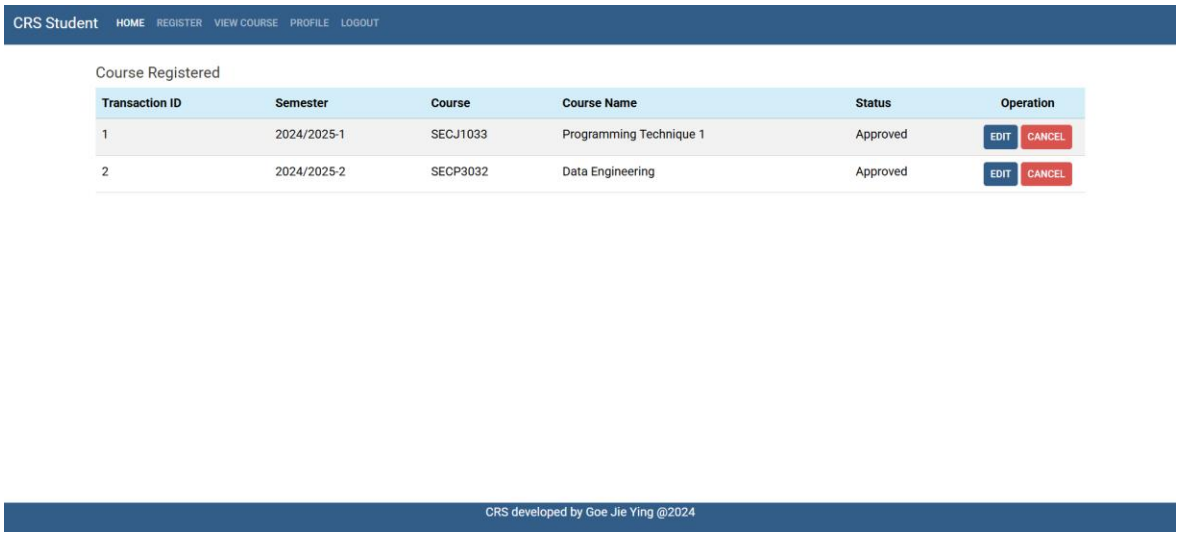
6.2 Student

6.2.1 Student Dashboard

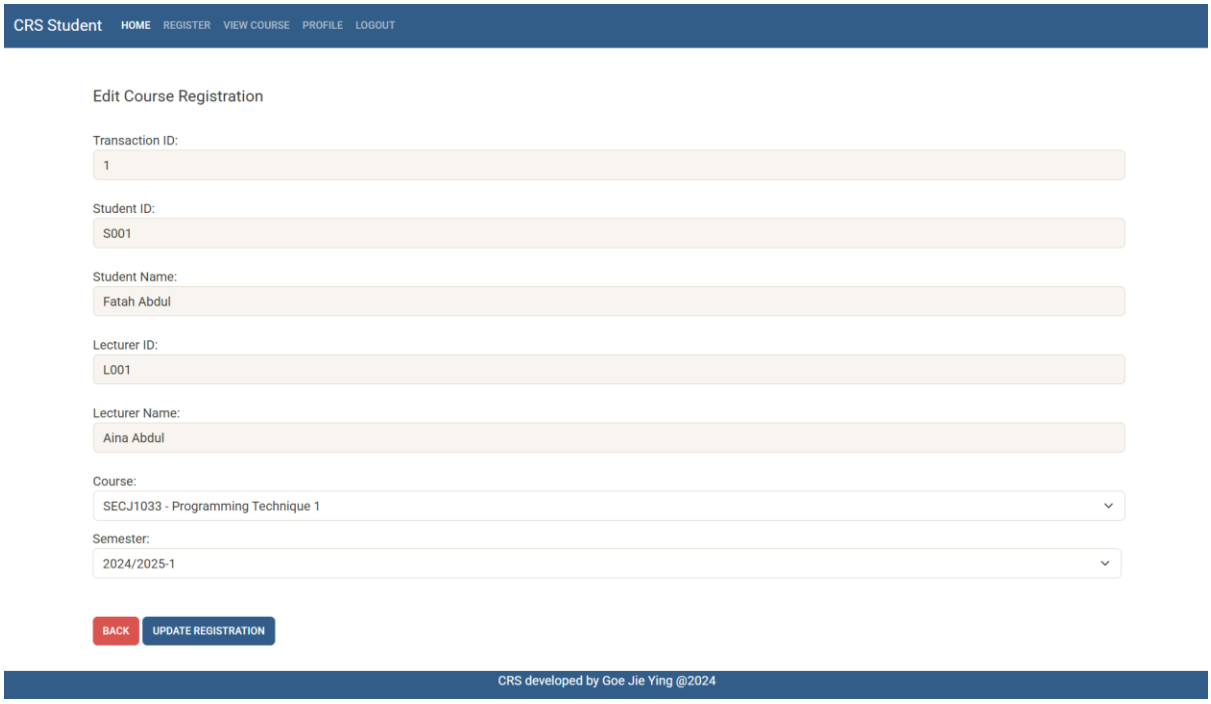
6.2.2 Student Register Course



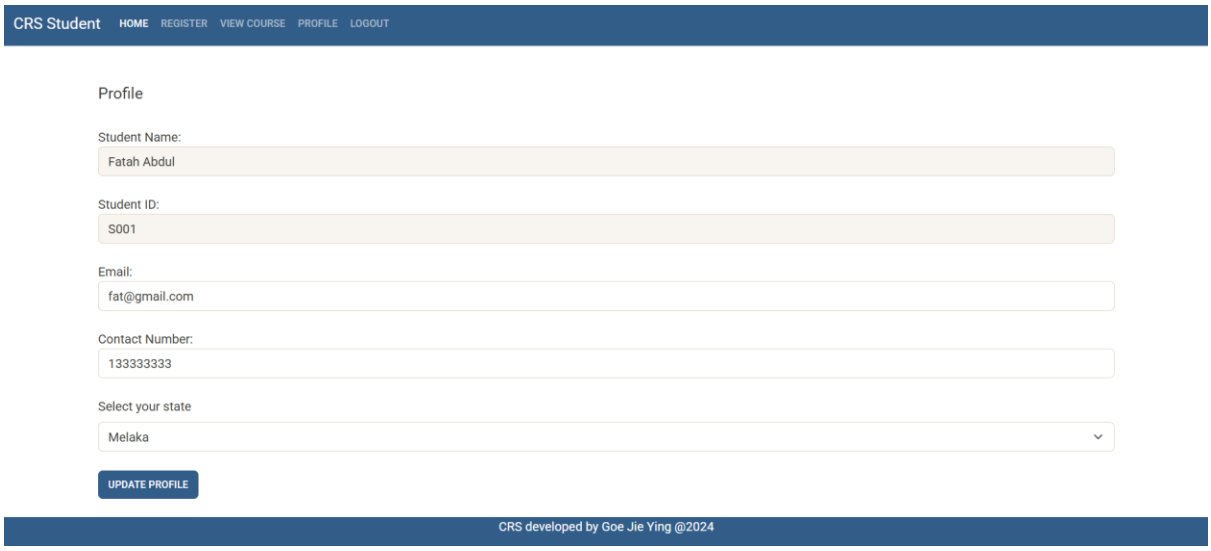
6.2.3 Student View Course



6.2.4 Student Edit Course Registration

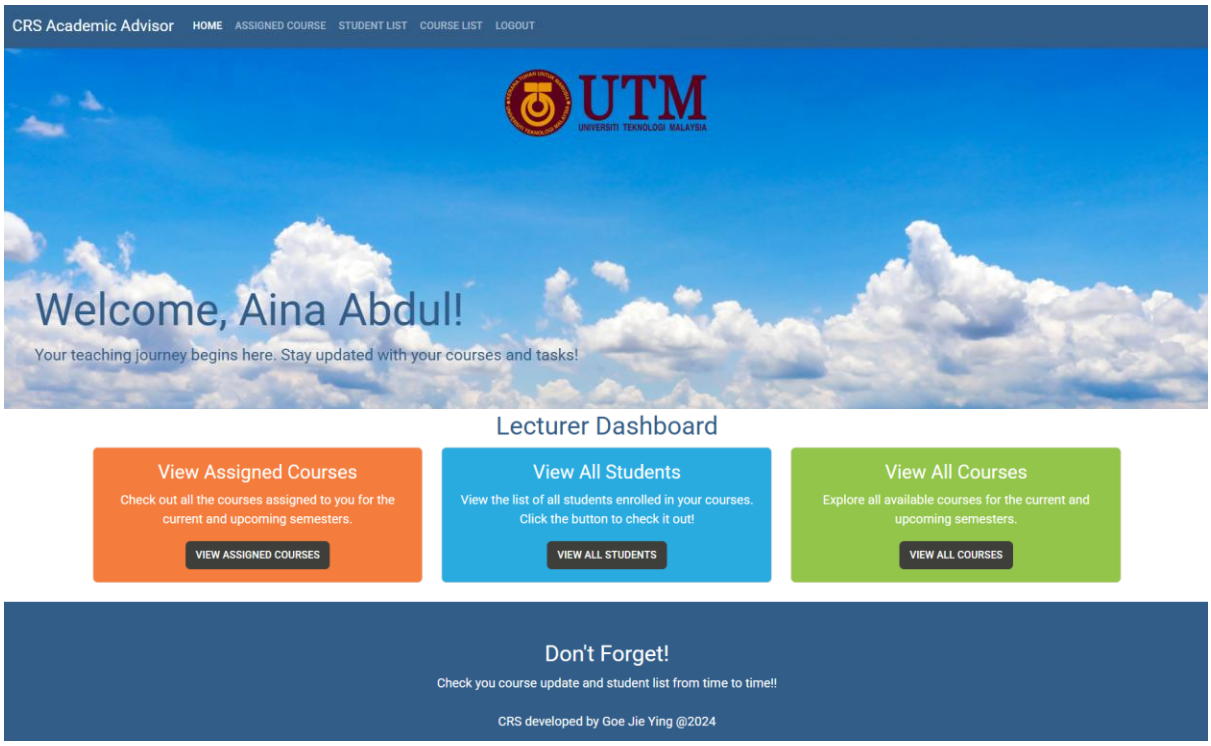


6.2.5 Student View and Edit Profile

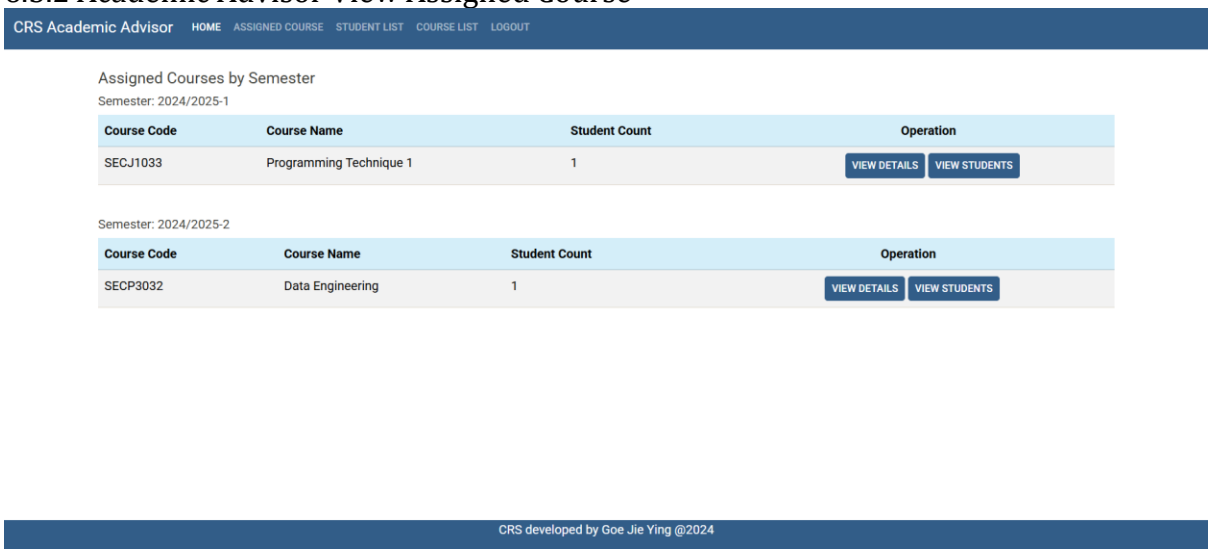


6.3 Academic Advisor (Lecturer)

6.3.1 Academic Advisor Dashboard



6.3.2 Academic Advisor View Assigned Course



6.3.3 Academic Advisor View Course Details

CRS Academic Advisor HOME ASSIGNED COURSE STUDENT LIST COURSE LIST LOGOUT

Course Details

Course Code	SECJ1033
Course Name	Programming Technique 1
Course Credit	3
Semester	2024/2025-1
Lecturer	L001
Max Students	4
Current Students	1

BACK

CRS developed by Goe Jie Ying @2024

6.3.4 Academic Advisor View Student List for Each Course

CRS Academic Advisor HOME ASSIGNED COURSE STUDENT LIST COURSE LIST LOGOUT

Students Enrolled in SECJ1033 - 2024/2025-1

Student ID	Student Name	Email
S001	Fatah Abdul	fat@gmail.com

CRS developed by Goe Jie Ying @2024

6.3.5 Academic Advisor View All Student List

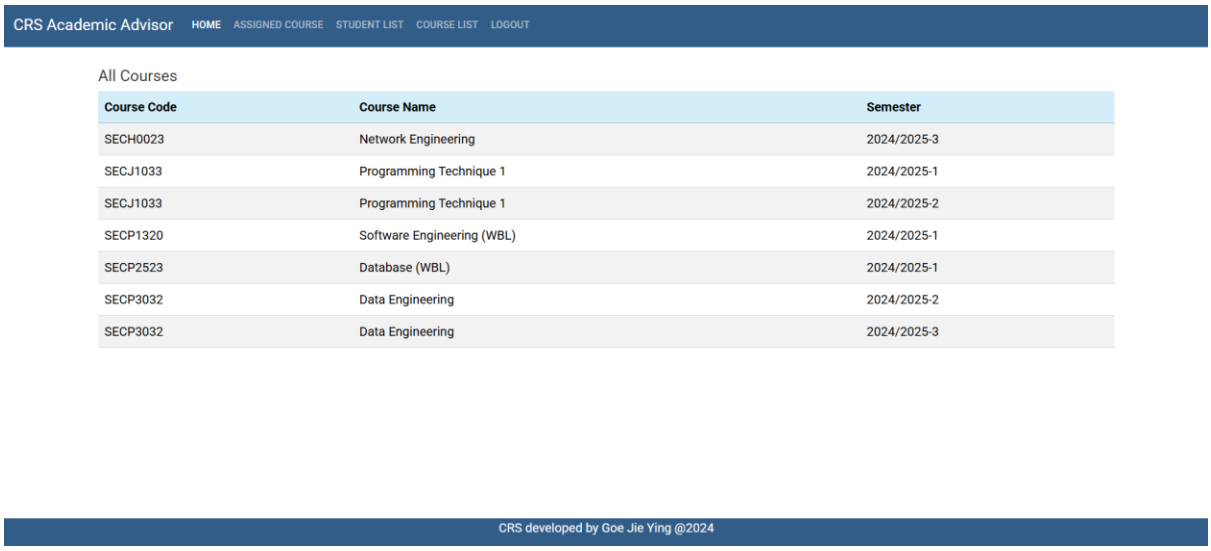
CRS Academic Advisor HOME ASSIGNED COURSE STUDENT LIST COURSE LIST LOGOUT

List of All Students

Student ID	Student Name	Email
S001	Fatah Abdul	fat@gmail.com
S002	Kamarul Abdul	kam@gmail.com
S003	Fatimah Abdul	fatimah@gmail.com
S004	Filza Abdul	filza@gmail.com
S005	Goe Jie Ying	goeying@graduate.utm.my

CRS developed by Goe Jie Ying @2024

6.3.6 Academic Advisor View All Course List



Course Code	Course Name	Semester
SECH0023	Network Engineering	2024/2025-3
SECJ1033	Programming Technique 1	2024/2025-1
SECJ1033	Programming Technique 1	2024/2025-2
SECP1320	Software Engineering (WBL)	2024/2025-1
SECP2523	Database (WBL)	2024/2025-1
SECP3032	Data Engineering	2024/2025-2
SECP3032	Data Engineering	2024/2025-3

6.4 Admin

6.4.1 Admin Dashboard

6.4.2 Admin View Course List

CRS Admin HOME COURSE LIST ADD COURSE STUDENT LIST REGISTRATION LIST LECTURER LIST ADD LECTURER LOGOUT

Course List

Course Code	Course Name	Course Credit	Semester	Lecturer ID	Lecturer	Maximum Students	Operation
SECH0023	Network Engineering	3	2024/2025-3	L006	Bullah Abdul	3	MODIFY DELETE
SECJ1033	Programming Technique 1	3	2024/2025-1	L001	Aina Abdul	4	MODIFY DELETE
SECJ1033	Programming Technique 1	3	2024/2025-2	L002	Fazura Abdul	2	MODIFY DELETE
SECP1320	Software Engineering (WBL)	3	2024/2025-1	L004	Abu Abdul	2	MODIFY DELETE
SECP2523	Database (WBL)	3	2024/2025-1	L002	Fazura Abdul	3	MODIFY DELETE
SECP3032	Data Engineering	3	2024/2025-2	L001	Aina Abdul	3	MODIFY DELETE
SECP3032	Data Engineering	3	2024/2025-3	L002	Fazura Abdul	3	MODIFY DELETE

CRS developed by Goe Jie Ying @2024

6.4.3 Admin Modify Course Detail

CRS Admin HOME COURSE LIST ADD COURSE STUDENT LIST REGISTRATION LIST LECTURER LIST ADD LECTURER LOGOUT

Modify Course

Course Code:
SECH0023

Course Name:
Network Engineering

Course Credit:
3

Lecturer:
Bullah Abdul

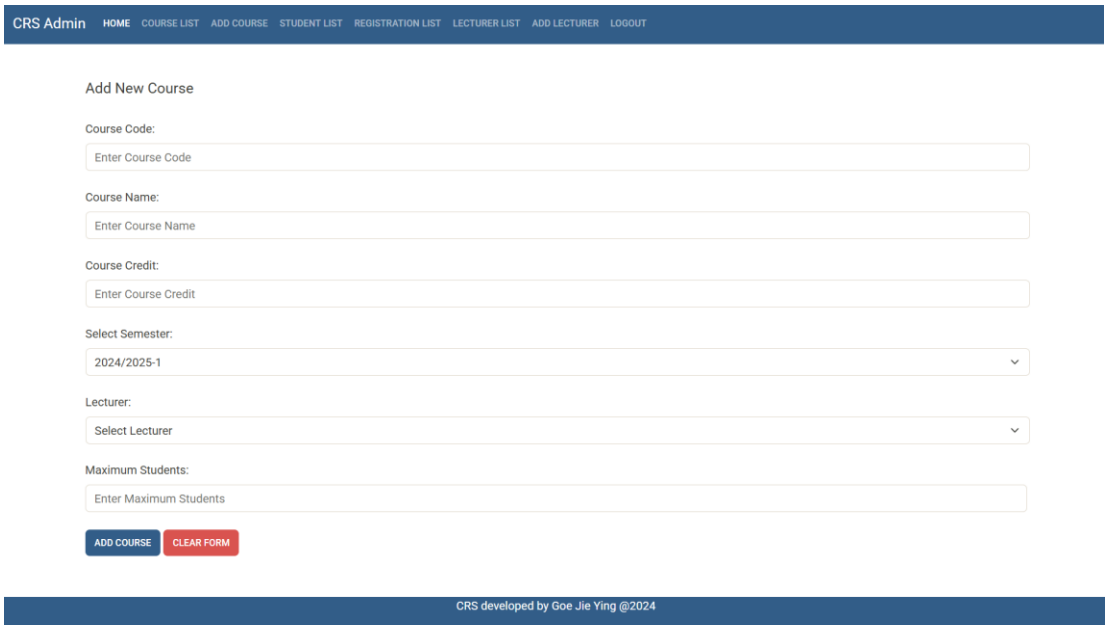
Semester:
2024/2025-3

Maximum Students:
3

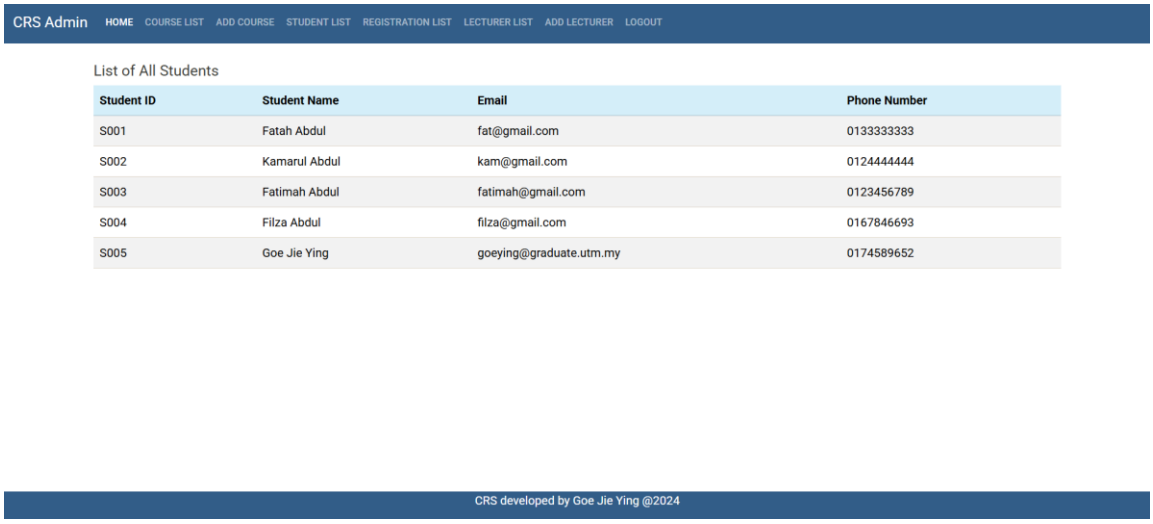
UPDATE COURSE CANCEL

CRS developed by Goe Jie Ying @2024

6.4.4 Admin Add New Course

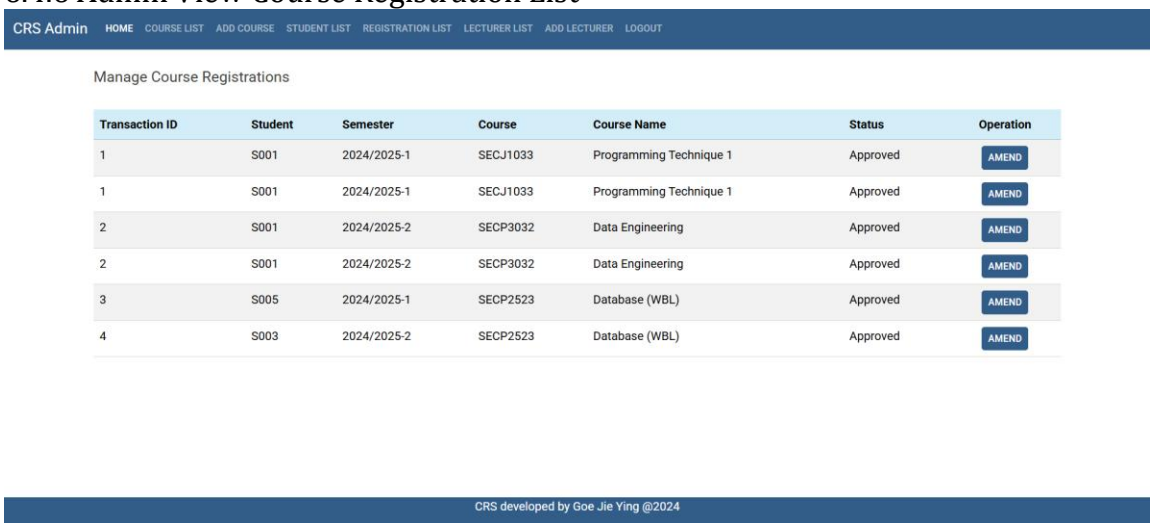


6.4.5 Admin View Student List



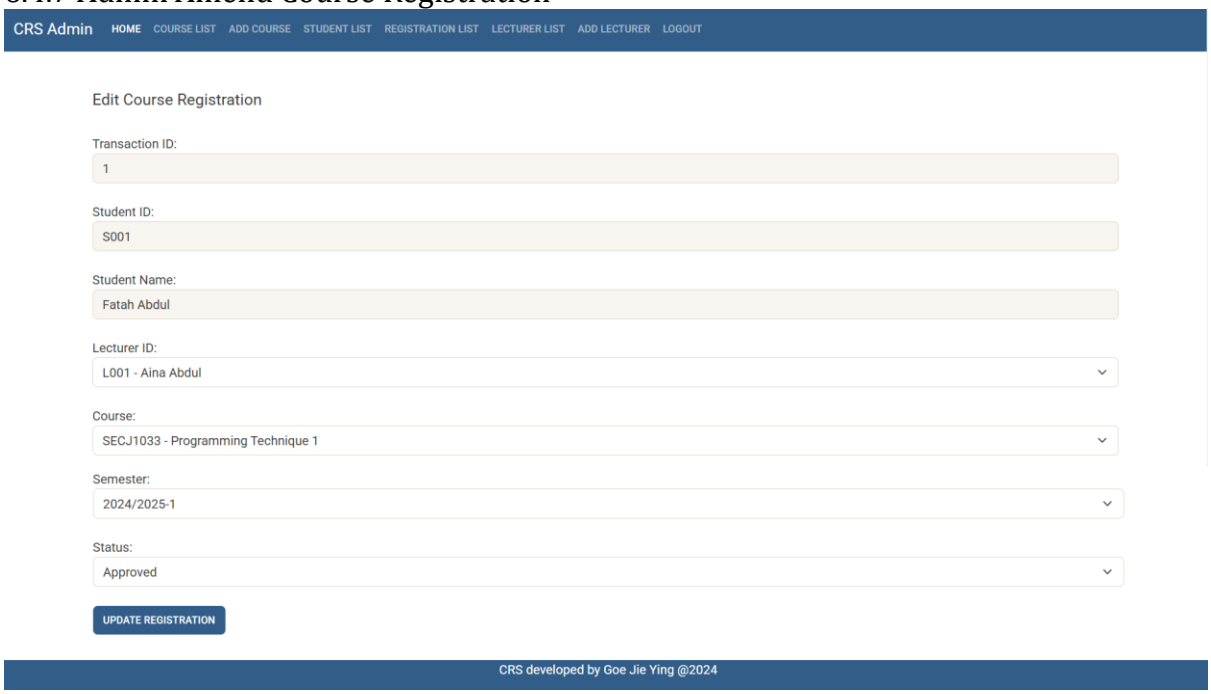
Student ID	Student Name	Email	Phone Number
S001	Fatah Abdul	fat@gmail.com	0133333333
S002	Kamarul Abdul	kam@gmail.com	0124444444
S003	Fatimah Abdul	fatimah@gmail.com	0123456789
S004	Filza Abdul	filza@gmail.com	0167846693
S005	Goe Jie Ying	goeying@graduate.utm.my	0174589652

6.4.6 Admin View Course Registration List

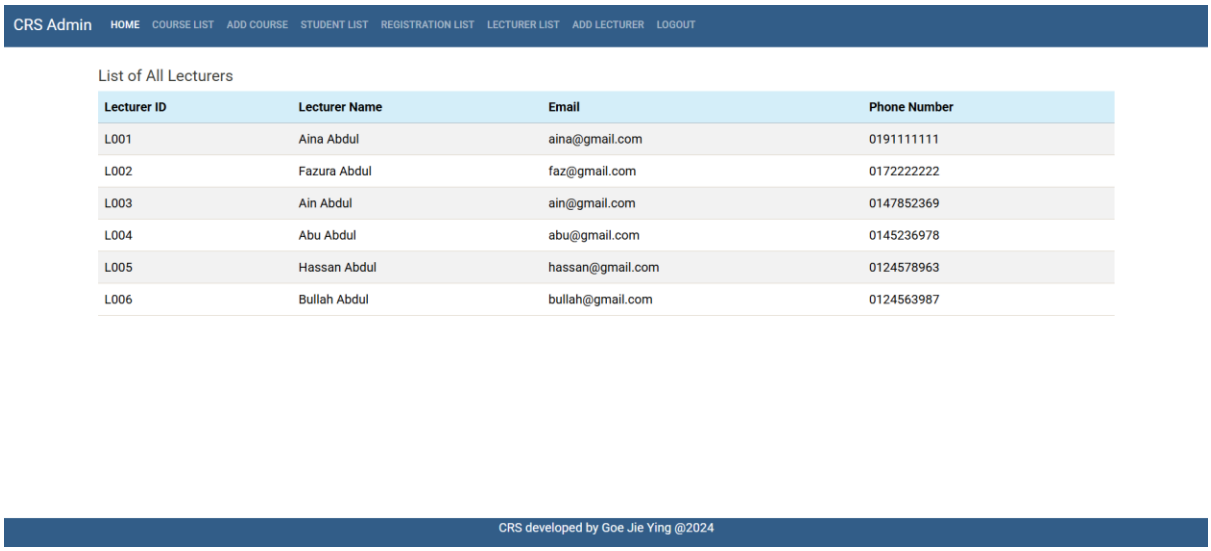


Transaction ID	Student	Semester	Course	Course Name	Status	Operation
1	S001	2024/2025-1	SECJ1033	Programming Technique 1	Approved	AMEND
1	S001	2024/2025-1	SECJ1033	Programming Technique 1	Approved	AMEND
2	S001	2024/2025-2	SECP3032	Data Engineering	Approved	AMEND
2	S001	2024/2025-2	SECP3032	Data Engineering	Approved	AMEND
3	S005	2024/2025-1	SECP2523	Database (WBL)	Approved	AMEND
4	S003	2024/2025-2	SECP2523	Database (WBL)	Approved	AMEND

6.4.7 Admin Amend Course Registration

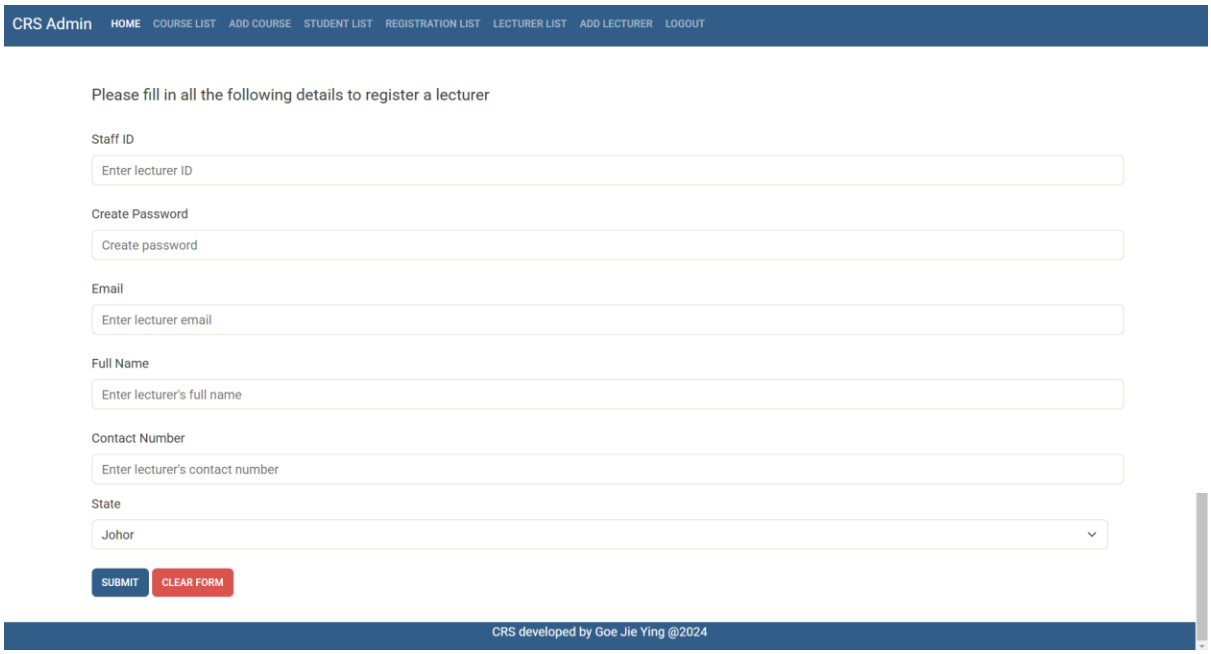


6.4.8 Admin View Lecturer List



Lecturer ID	Lecturer Name	Email	Phone Number
L001	Aina Abdul	aina@gmail.com	0191111111
L002	Fazura Abdul	faz@gmail.com	0172222222
L003	Ain Abdul	ain@gmail.com	0147852369
L004	Abu Abdul	abu@gmail.com	0145236978
L005	Hassan Abdul	hassan@gmail.com	0124578963
L006	Bullah Abdul	bullah@gmail.com	0124563987

6.4.9 Add Lecturer



7.0 Localhost Setup

1. Open XAMPP, start Apache and MySQL.
2. Open browser and key in the link below in search bar.

Link: localhost/crs/
Database Name: db_crs

8.0 Hosting URL

https://crsystem.great-site.net/

9.0 User Credential

ID	Password
Student	
S001 – S004	123456
Lecturer	
L001 & L002	123456
Admin	
I001 & I002	123456